Saturation Defects and Price Dependence
in Synchronized Optimal Farey Parent Words

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Abstract

For two genuine, independently optimized rank-three Farey parent words, we study the signed discrepancy letters obtained by synchronized cyclic differentiation. We distinguish the integral span of the actual letters from the translation-invariant span of their differences. A generalized CRT decomposition gives a source-linear generator presentation for the latter without expanding the joint period. For any primitive daughter vector, the associated finite quotient determines the *exact* attainable determinant ideal and thus the precise failure of parent-difference basis completion. Explicit positive unimodular infinite families show that the raw step lattice can have unbounded index and that the difference lattice can have rank one even when the child has an essential new optimal letter. In another genuine family a single affine letter restores the complete daughter lattice from an index-29 difference lattice; an exact price-chamber theorem classifies the global minimizing positive primitive rays, including three ties at a wall. The entire source alphabet and daughter lattice are constant across this wall although the winning ray changes, ruling out source-alphabet-only selection without numerical price data. The unrestricted original-parent-to-child optimal-word transfer problem remains open.

Keywords. Farey parent words; optimal Parikh fibers; Chinese remainder theorem; lattice saturation; quotient group; determinant defect. **Preprint status.** This manuscript gives self-contained proofs for the stated exact arithmetic claims and concrete families. The underlying CRT, lattice-index and quotient arguments are classical; the Farey-optimal hypotheses and genuine infinite obstruction families provide the paper-specific mathematical content. The final external priority review remains independent of mathematical correctness.

Introduction

The usual synchronization question asks when compatible cyclic word structures can be assembled; the arithmetic question considered here is different. We begin with two *already grade-optimal* collections of original Parikh records arising from a single positive unimodular Farey frame. Their synchronized differences are authentic source data, but there is no a priori reason that these differences generate the saturated lattice needed to describe a new child's optimum. The central issue is therefore not the CRT alone: it is how much integral information survives passing from absolute source letters to their translation-invariant differences, and whether a primitive child direction admits a determinant-minimal complementary source difference.

The results below give both an exact quantitative defect formula and infinite genuine optimized-word obstructions. The abstract group arguments are elementary; the nontrivial source-specific content is the existence of these phenomena with complete optimal parent provenance and, in the index-29 family, a precise price-dependent repair. The latter also shows why retaining the full

source letter alphabet without the numerical price cannot determine the new optimal positive ray. Existing Christoffel synchronization work is compared after the proofs; the article does not assert an unrestricted transfer theorem.

1 Genuine source and CRT synchronization

Let $B \in SL_3(\mathbb{Z})$ have strictly positive integer entries and put $P = B\mathbf{1}$, $Q = Be_j$, and $C = P + Q = (p, q, s)$. For a positive source $R = (p_R, q_R, s_R)$ and a fundamental target $0 \le r < p_R$, an *original physical Parikh record* is a triple $z = (x, u, v) \in \mathbb{Z}^3_{\ge 0}$ satisfying

$$A_R z = p_R x - q_R u - s_R v = r.$$

Its *grade* is $N(z) = x + u + v$. The *complete optimum fiber* at r consists of all grade-minimizing physical records; the complete target-indexed WORD records all these fibers, including ties. Equivalently, minimizing the grade at a fixed r is minimizing the positive defect price

$$(p_R + q_R)u + (p_R + s_R)v = p_R N(z) - r.$$

The two original parent WORDs for P and Q are optimized *independently*, before their Farey sum is formed. The complete optimum fiber is retained as the underlying semantic datum; the cyclic derivative alphabet below is formed from *one consistently selected anchor per residue*, not from the union of differences among every tied optimum. Fix a deterministic lexicographic ordering of the three physical coordinates to select one anchor $z_R(r)$ in each optimum fiber; the general algebraic statements hold for any consistently selected anchors. Denote the parent first coordinates by $c = P_0$ and $w = Q_0$. Each anchor has the canonical periodic extension $z_R(r + p_R) = z_R(r) + e_x$. Write

$$d_R(i) = z_R(i + 1) - z_R(i), \qquad i \pmod{p_R}, \quad R = P, Q.$$

Put $D = \mathrm{diag}(1, -1, -1)$ and interpret the source residual row as $A_R = R^T D$. Since B is unimodular, a primitive simultaneous source normalizer is explicitly $v_0 = DB^{-T}e_j$, which satisfies $A_P v_0 = A_Q v_0 = 1$. The integral parent transporter $T = I - v_0(A_P - A_Q)$ is unimodular: $A_P T = A_Q$ and $\det T = 1 - (A_P - A_Q)v_0 = 1$. Choose any primitive integral basis K_P of $\ker_{\mathbb{Z}} A_P$ (one canonical choice is $DB^{-T}(e_1 - e_3, e_2 - e_3)$); the parent discrepancy coordinates are $\mu(r) = K_P^{-1}(Tz_Q(r) - z_P(r)) \in \mathbb{Z}^2$.

The identification with the daughter lattice also has an explicit unimodular origin. Choose any $j' \ne j$ and set $U = I + e_j e_{j'}^T$, so $U\mathbf{1} = \mathbf{1} + e_j$ and $\det U = 1$. Then $S = DB^{-T}U^{-T}B^T D \in SL_3(\mathbb{Z})$ satisfies $A_C S = A_P$. Consequently S identifies the primitive parent kernel with the primitive daughter kernel, and projection of SK_P onto the (u, v)-coordinates is an integral basis F of $\Lambda(C) = \{(u, v) \in \mathbb{Z}^2 : qu + sv \equiv 0 \pmod{p}\}$, with $|\det F| = p$. The source letters are thus mapped to daughter physical vectors by an explicitly integral lattice isomorphism, not by an assumed real-linear identification.

Lemma 1.1 (Authentic CRT discrepancy)**.** *For all integers r, the synchronized discrepancy increment is*

$$\delta_r = \mu(r + 1) - \mu(r) = K_P^{-1}(Td_Q(r \bmod w) - d_P(r \bmod c)).$$

Set $g = \gcd(c, w)$ and $L = \mathrm{lcm}(c, w)$. Over one full period $0 \le r < L$, the ordered pair $(r \bmod c, r \bmod w)$ visits each (i, j) with $i \equiv j \pmod{g}$ exactly once and no other pair. Thus the actual physical discrepancy alphabet is precisely $\mathcal{D} = \{\delta_{ij} : i \equiv j \pmod{g}\}$; all letters are source-derived, not hypothetical daughter optimum vectors.

Proof. Subtract the defining equations for $\mu(r+1)$ and $\mu(r)$ and use periodicity of both optimized parent derivatives. The generalized CRT has a solution for $r \equiv i \pmod{c}$, $r \equiv j \pmod{w}$ exactly when $i \equiv j \pmod{g}$ and then a unique solution modulo L. □

On transporting these coordinates through the known full daughter lattice basis, each δ_{ij} is in the integral daughter lattice $\Lambda = \{(x, y) \in \mathbb{Z}^2 : qx + sy \equiv 0 \pmod{p}\}$ of determinant p. The letters may be written in the primitive parent-kernel coordinate lattice $\mathbb{Z}^2$ or, after the integral source-carried basis identification, as physical vectors in Λ; ranks and subgroup indices are unchanged by this lattice isomorphism. We specify the chosen coordinates in each explicit family.

2 Compressed raw and difference modules

The *raw* parent letter module and its *translation-invariant* difference module are, respectively,

$$L_{\mathrm{raw}} = \langle \delta : \delta \in \mathcal{D} \rangle_{\mathbb{Z}}, \quad L_{\mathrm{diff}} = \langle \delta - \delta' : \delta, \delta' \in \mathcal{D} \rangle_{\mathbb{Z}}.$$

Fix an anchor (c_0, c_0) in every CRT congruence class $c_0 \in \{0, \dots, g-1\}$; denote $a_{c_0} = \delta_{c_0,c_0}$ and put

$$U_i = \delta_{i,c_0} - a_{c_0}, \qquad V_j = \delta_{c_0,j} - a_{c_0} \quad (i \equiv j \equiv c_0 \pmod{g}).$$

Theorem 2.1 (Exact source-sized generator theorem)**.** *The complete full-period translation-invariant discrepancy module satisfies*

$$\boxed{L_{\mathrm{diff}} = \langle U_i, V_j, a_{c_0} - a_0 : \ i, j \ \textit{eligible}, \ 1 \le c_0 < g \rangle_{\mathbb{Z}}.}$$

At most $c + w - g - 1$ nonzero parent-derived integral difference vectors suffice. Hence the rank, Hermite normal form, and exact index (if full rank) can be found using a linear number of integer gcd operations in the two-parent word input size, without expanding L.

Proof. For each class the linearity of K_P^{-1} and the separability of the two parent derivatives imply $\delta_{ij} = a_{c_0} + U_i + V_j$. Every displayed generator is the difference of two actually attainable CRT phase letters. Conversely subtracting two arbitrary letters from classes c_0 and d_0 gives $(a_{c_0} - a_{d_0}) + U_i + V_j - U_{i'} - V_{j'}$, in the subgroup generated on the right. This proves equality. Each of the g classes contributes one zero U and one zero V at its anchor, so there are at most $(c - g) + (w - g) + (g - 1) = c + w - g - 1$ nonzero listed generators. In rank two, iterative extended gcd produces HNF without ever forming the full CRT letter list. □

Theorem 2.2 (The affine origin and the exact raw-module recovery)**.** *For any finite nonempty integral alphabet $\mathcal{D} \subseteq \Lambda$, choose $a_0 \in \mathcal{D}$. Then*

$$L_{\mathrm{raw}} = L_{\mathrm{diff}} + \mathbb{Z}a_0, \qquad L_{\mathrm{raw}}/L_{\mathrm{diff}} = \langle [a_0] \rangle$$

is cyclic (possibly infinite). In particular, $\mathrm{rank}(L_{\mathrm{raw}}) - \mathrm{rank}(L_{\mathrm{diff}}) \in \{0, 1\}$*. If L_{diff} has rank two and $m = [\Lambda : L_{\mathrm{diff}}]$, then*

$$\boxed{[\Lambda : L_{\mathrm{raw}}] = \frac{m}{\mathrm{ord}_{\Lambda/L_{\mathrm{diff}}}([a_0])}.}$$

Thus one genuine affine anchor supplies exactly the missing cyclic coset of the difference module, and its order measures how much saturation the absolute origin restores.

Proof. Every $a \in \mathcal{D}$ is $(a - a_0) + a_0$ and $a - a_0 \in L_{\mathrm{diff}}$, giving both inclusions for L_{raw}. Therefore the raw quotient is generated by the image of a_0. If the difference module is full rank, use

$$\Lambda/L_{\mathrm{raw}} \cong (\Lambda/L_{\mathrm{diff}})/\langle [a_0] \rangle$$

and count group orders. The rank assertion follows because adjoining one generator raises the rational rank by at most one. □

Remark 2.3. *Theorem 2.2 is an elementary group extension, not a novel abstract-group theorem. Its application prevents conflating the signed span of actual parent letters with their translation-invariant differences. The exact quotient of a rank-one difference module by the whole child lattice is not finite, so the finite-order formula cannot be used in that case.*

3 Exact determinant quantization

For any rank-two integral lattice Λ of positive determinant p, every vector z is called Λ-primitive if $\Lambda/\mathbb{Z}z$ is torsion-free (equivalently, z extends to an integral basis of Λ).

Theorem 3.1 (Sharp quotient-order determinant defect)**.** *Let $L \subseteq \Lambda$ be rank two with $m = [\Lambda : L]$, and let $z \in \Lambda$ be Λ-primitive. Put $G = \Lambda/L$ and $o = \mathrm{ord}_G([z])$. Then*

$$\boxed{\det(z, L) = \{\det(z, h) : h \in L\} = p\frac{m}{o}\mathbb{Z}.}$$

In particular the least nonzero determinant attainable with an L-vector is exactly pm/o, and a full determinant-p complement $h \in L$ exists if and only if $[z]$ generates G.

Proof. Primitivity of z gives a basis (z, h_0) of Λ, with $|\det(z, h_0)| = p$. Consequently $\phi_z : \Lambda \to \mathbb{Z}$, $h \mapsto \det(z, h)/p$, is surjective and has kernel $\mathbb{Z}z$. Write $\phi_z(L) = d\mathbb{Z}$ for some $d \geq 1$. By the third isomorphism theorem,

$$\mathbb{Z}/d\mathbb{Z} \simeq \Lambda/(L + \mathbb{Z}z) \simeq (\Lambda/L)/\langle [z] \rangle,$$

whose order equals m/o. Therefore $d = m/o$. All multiples occur because $\phi_z(L)$ is exactly this ideal. Multiplying by p proves the formula and the determinant-p equivalence. □

Corollary 3.2 (HNF and invariant factors)**.** *If L has physical HNF basis $(a, 0), (b, d)$, then for every $z = (x, y)$*

$$\gcd(a|y|, |dx - by|) = \gcd\{|\det(z, h)| : h \in L\}.$$

If z is Λ-primitive, this number equals pm/o. Further, if $G \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z}$ with $d_1 \mid d_2$ and $d_1 > 1$, it is at least pd_1 for every Λ-primitive z. Thus a noncyclic quotient, if it arises in the relevant source category, blocks every *determinant-p complement chosen from parent-letter differences.*

Proof. The integer ideal generated by determinants of z with the two HNF generators has the displayed positive gcd. For the second assertion, $\mathrm{ord}([z]) \leq d_2$, while $m = d_1 d_2$, and Theorem 3.1 gives the inequality. □

Example 3.3 (Eightfold defect on a genuine Farey edge)**.** *An authentic positive determinant-one frame is*

$$B = \begin{pmatrix} 3 & 6 & 4 \\ 2 & 5 & 3 \\ 8 & 19 & 12 \end{pmatrix}, \quad C = (16, 12, 47).$$

Here the actual parents are $P = (13, 10, 39)$ and $Q = (3, 2, 8)$. Their independently optimized fundamental words are

$$z_P(r) = ((r + 10u_r)/13, u_r, 0), \quad u_r = [9r]_{13}, \qquad z_Q(r) = (r, r, 0) \quad (0 \leq r < 3).$$

For P, the v-coefficient has zero residue modulo 13 and strictly positive price, so $v = 0$ at any optimum; the least u solving $10u \equiv -r \pmod{13}$ is $[9r]_{13}$. For Q, $u + v \equiv r \pmod{3}$ and its u price is 5, versus v price 11, making $(r, r, 0)$ optimal for $0 \leq r < 3$. Thus the actual cyclic derivative alphabets are $\{(7, 9, 0), (-3, -4, 0)\}$ and $\{(1, 1, 0), (-1, -2, 0)\}$. Take $v_0 = (3, -4, 2)^T$, so $A_P v_0 = A_Q v_0 = 1$, and the genuine integer transporter $T = I - v_0(A_P - A_Q)$. In the primitive parent kernel basis $K_P = ((10, 13, 0)^T, (3, 0, 1)^T)$ the compatible original-step CRT calculation gives four physical daughter letters

$$\mathcal{D} = \{(-19, -4), (10, -8), (-16, 0), (13, -4)\}.$$

Their pairwise differences generate exactly

$$L_{\mathrm{diff}} = \langle (32,0),(3,4)\rangle_{\mathbb{Z}}, \qquad [\Lambda : L_{\mathrm{diff}}] = 128/16 = 8.$$

For example, $((29,-4),(4,0))$ *is a full daughter lattice basis of determinant* 16 *giving the same physical alphabet. The primitive positive daughter winner* $z_\star = (3,4)$ *has a short certificate within the paper. For* $C = (16,12,47)$*, every strictly positive feasible pair satisfies* $v \equiv 12u \pmod{16}$*, with price* $\Phi(u,v) = 28u + 63v$*. For* $1 \le u \le 9$*, the least positive admissible values of* v *are, respectively,* $12,8,4,16,12,8,4,16,12$*; direct substitution shows the unique minimum in this range is* $(u,v) = (3,4)$*, of price* 336*. For* $u \ge 10$ *and* $v \ge 1$*, the price is at least* $28 \cdot 10 + 63 = 343 > 336$*. Thus* $z_\star$ *is the unique strictly positive price minimizer, independently of any finite computational enumeration. It is primitive in the full daughter lattice (in the displayed full basis it has coordinates* $(-1,8)$*) and belongs to the difference lattice. Hence its quotient class has order one, and the minimum nonzero determinant with any parent difference is* $16 \cdot 8 = 128$*; indeed* $\gcd(128,0) = 128$*. In contrast,* $z' = (4,0)$ *is primitive in* Λ*, has quotient order eight, and admits determinant 16. This dependence on the quotient class is invisible to the subgroup index alone.*

4 Two authentic infinite obstructions

The next results are derived from fully optimized genuine parent words rather than arbitrarily assigned alphabet vectors. We include the original-parent optimization and exact transporter calculations, so no daughter-optimum oracle or external research note is needed for these proofs.

Theorem 4.1 (Unbounded raw index but rank-one differences)**.** *For every odd* $n \ge 1$*, let* $p = n + 2$ *and* $k = 4p$*, and set*

$$B_n = \begin{pmatrix} p-2 & 1 & 1 \\ 2p-5+k(p-2) & 2+k & 2+k \\ 2p-5 & 1 & 2 \end{pmatrix} \in SL_3(\mathbb{Z}).$$

Use its authentic second-column parent Q *and the original sum parent* $P = B_n\mathbf{1}$*. Then the complete original parent optimum words give the source-kernel discrepancy alphabet*

$$\mathcal{D}_n = \left\{ \left(p-1, \frac{p-3}{2}\right), \left(p-1, \frac{p-1}{2}\right) \right\}.$$

Both types are attained by genuine CRT phases. Consequently

$$\operatorname{rank} L_{\mathrm{raw},n} = 2, \quad [\mathbb{Z}^2 : L_{\mathrm{raw},n}] = p - 1 = n + 1 \longrightarrow \infty, \quad L_{\mathrm{diff},n} = \mathbb{Z}(0,1).$$

The latter has rank one, despite the raw module having rank two.

Proof. The displayed matrix is strictly positive and has determinant one by direct expansion (equivalently, integer elementary column and row operations reduce it to the identity frame). Its two parents are

$$P = (p,\ (k+2)p - 1,\ 2p-2), \qquad Q = (1, k+2, 1).$$

As the fundamental period of Q is one, its sole original optimum mark is zero and its cyclic step is $d_Q = e_x$. The first coordinate p of P is odd. Since $s_P = 2p - 2 \equiv -2 \pmod{p}$, for each $0 \le r < p$ the *unique least* nonnegative v with a $u = 0$ feasible mark is

$$v_r = [r \cdot 2^{-1}]_p, \quad z_P(r) = \left(\frac{r + (2p-2)v_r}{p},\ 0,\ v_r \right).$$

This is genuinely optimal, not a formally chosen parent shadow: the price of one u is $a_P = p + q_P = (k+3)p - 1$, while the greatest candidate $u = 0$ price is at most $(p-1)(3p-2)$. For $k = 4p$ their difference is

$$(k+3)p - 1 - (p-1)(3p-2) = p^2 + 8p - 3 > 0.$$

Hence every parental optimum has $u = 0$, and v_r is minimal because each extra p of v strictly raises the cost. Its cyclic differences satisfy

$$\Delta v_r = v_{r+1} - v_r \in \left\{ \frac{p+1}{2}, -\frac{p-1}{2} \right\}, \quad d_P(r) = \left(\frac{1 + (2p-2)\Delta v_r}{p}, 0, \Delta v_r \right);$$

both possibilities occur as the residues v_r form a complete permutation.

We now compute the synchronized *original-parent* transport, rather than choosing daughter rays. Write $A_R = (p_R, -q_R, -s_R)$, take $v = (-k, -1, 1)^T$ (so $A_P v = 1$), and set

$$T = I - v(A_P - A_Q), \qquad A_P T = A_Q.$$

It follows that $Td_Q = Te_x = (1 + k(p-1), p-1, -(p-1))^T$. A primitive parent kernel basis is

$$K_P = (\kappa_1, \kappa_2), \quad \kappa_1 = (k + 2 - p, 1, -(p+1)/2)^T, \quad \kappa_2 = (2p-2, 0, p)^T.$$

Indeed $A_P\kappa_1 = A_P\kappa_2 = 0$ and their (u, v)-projections have determinant p, the index of the projected kernel congruence; hence they span the full integer parent kernel. Directly solving $K_P\delta = Td_Q - d_P(r)$ gives

$$\delta_r = \begin{cases} (p-1, (p-3)/2)^T, & \Delta v_r = (p+1)/2, \\ (p-1, (p-1)/2)^T, & \Delta v_r = -(p-1)/2. \end{cases}$$

The two vectors differ by $(0, 1)$ and have determinant $p - 1$, proving all the stated raw and difference module assertions, with no appeal to a child optimum computation. □

Theorem 4.2 (Rank-one source but a compulsory new daughter letter)**.** *For odd $n \geq 1$, the same $p = n + 2$, $k = 4p$, and the alternative authentic frame*

$$B'_n = \begin{pmatrix} p-2 & 1 & 1 \\ 2p - 5 + k(p-2) & 2+k & 2+k \\ p-2 & 1 & 2 \end{pmatrix} \in SL_3(\mathbb{Z})$$

with third-column *insertion has two independently optimized parents with $u = 0$ throughout their fundamental words. Nevertheless its entire original-parent discrepancy-step alphabet is only $\mathcal{D}'_n = \{(0,0), (0,1)\}$ in primitive kernel coordinates, whereas the genuine child*

$$C = (p+1,\ 2p + 1 + k(p+1),\ p+3)$$

has exactly one optimum record at every target $0 \leq r < p + 1$, namely

$$u_r = r \bmod 2, \quad v_r = \left[-\frac{r - u_r}{2} \right]_{(p+1)/2}, \quad x_r = \frac{r + q_C u_r + s_C v_r}{p+1}.$$

In particular, $u_r = 1$ is compulsory at every odd child target even though the parent optimum words contain only $u = 0$.

Proof. Both strict positivity and determinant one follow by expansion. The original parents are

$$P = (p, (k+2)p - 1, p+1), \qquad Q = (1, k+2, 2).$$

Again the only fundamental Q mark is zero. Because $s_P \equiv 1 \pmod p$, its $u = 0$ optimum at target r is $v_r = [-r]_p$ with $z_P(r) = ((r + (p+1)v_r)/p, 0, v_r)$. Its price is at most $(p-1)(2p+1)$, whereas any $u \geq 1$ has price at least $(k+3)p - 1$; their difference at $k = 4p$ is $2p(p+2) > 0$. Thus these are exactly all parent optima.

The cyclic derivative of this complete parental word is

$$d_P(0) = (p, 0, p-1)^T, \qquad d_P(r) = (-1, 0, -1)^T \quad (1 \leq r < p).$$

Indeed $z_P(0) = (0, 0, 0)$, while $z_P(1) = (p, 0, p-1)$; for every subsequent original phase (including the cyclic last-to-first extension) the difference is $(-1, 0, -1)$. The exact positions matter for the ordered WORD, although they do not affect the two-letter set. Take the genuine integral source transport $v = (-1, 0, -1)^T$, which satisfies $A_P v = 1$, and $T = I - v(A_P - A_Q)$. Hence $Td_Q = Te_x = (p, 0, p-1)^T$. One primitive parent kernel basis is

$$K_P = \big((k+3, 1, 1)^T,\ (p+1, 0, p)^T\big).$$

The difference $Td_Q - d_P$ is either the zero vector or its second basis column; hence the *complete genuine* phase alphabet is $(0, 0), (0, 1)$, each type occurring.

For the child write $N = p + 1$, an even number, and observe $q_C \equiv -1 \pmod N$ and $s_C \equiv 2 \pmod N$. Thus a feasible pair of nonnegative letter counts satisfies $u - 2v \equiv r \pmod N$, forcing $u \equiv r \pmod 2$. Choosing the least nonnegative $u_r \in \{0, 1\}$ of this parity and the displayed least $v_r \in [0, N/2 - 1]$ solves the congruence and gives a nonnegative integer x_r. Let $a_C = N + q_C$, $b_C = N + s_C$; the chosen $k = 4p$ gives

$$a_C = 4p^2 + 7p + 2, \quad b_C = 2p + 4, \quad a_C - (N-1)b_C = 2p^2 + 3p + 2 > 0.$$

Increasing u by two incurs at least $2a_C$, more than any possible saving relative to $v_r b_C \leq (N/2 - 1)b_C$. At the fixed minimal $u = u_r$, all other solutions have $v = v_r + jN/2$, $j \geq 0$, and strictly greater price for $j > 0$. Therefore the stated mark is the unique actual daughter optimum, with $u_r = 1$ on odd targets. No unspecified child search or generic optimization theorem is required. □

5 A genuine fixed-index family with a complete price-wall repair

The preceding obstructions are not impossibility statements about source transfer with access to absolute parent-basis data. We make this distinction exact on an independently optimized infinite family.

Theorem 5.1 (Index 29 and three exact minimizing rays at the wall)**.** *For every integer $t \geq 3$, let*

$$B_t = \begin{pmatrix} 3 & 6 & 4 \\ 2+3t & 5+6t & 3+4t \\ 8 & 19 & 12 \end{pmatrix} \in SL_3(\mathbb{Z})$$

with insertion of the first column. Its genuine parents are $P_t = (13, 10 + 13t, 39)$ and $Q_t = (3, 2 + 3t, 8)$, its child is $C_t = (16, 12 + 16t, 47)$, and its complete original-parent CRT difference lattice has index

$$\boxed{[\Lambda(C_t) : L_{\mathrm{diff},t}] = 29, \qquad L_{\mathrm{raw},t} = \Lambda(C_t).}$$

A full source-frame-defined child lattice basis is

$$f_1 = (29, -4), \qquad f_2 = (4, 0), \qquad \det(f_1, f_2) = 16.$$

Define $z_A = -f_1 + 8f_2 = (3,4)$ *and* $z_B = -3f_1 + 22f_2 = (1,12)$. *For the strictly positive primitive child-ray price* $\Phi_t(x,y) = (28+16t)x + 63y$, *the complete set of positive price-minimizing primitive lattice rays is*

$$\mathcal{M}_t = \begin{cases} \{(3,4)\}, & 3 \le t \le 13, \\ \{(1,12),(2,8),(3,4)\}, & t = 14, \\ \{(1,12)\}, & t \ge 15. \end{cases}$$

At $t = 14$ *the canonical lex-first ray is* $(1,12)$. *Moreover,* the full daughter lattice and the complete signed physical CRT letter alphabet are identical for all $t \ge 3$, *even though the optimal ray changes with* t.

Proof. All entries of B_t are positive and its determinant equals one. We first verify that the parent records really are independently optimum. For P_t, the fundamental constraint modulo 13 is $10u \equiv -r \pmod{13}$, so

$$u_r = [9r]_{13}, \quad v_r = 0, \quad x_r = tu_r + \frac{r + 10u_r}{13}, \qquad 0 \le r < 13$$

is the unique minimum. Indeed every admissible u is $u_r + 13j$ for $j \ge 0$, and $v \ge 0$ contributes an extra $4v$ to the physical grade, while increasing u adds a positive multiple to it. For Q_t the fundamental congruence is $u + v \equiv r \pmod 3$; the unique optimum is $(x,u,v) = (3r,0,r)$ for $r = 0,1,2$. This follows by comparing the least representatives $u + v = r$ and then observing that each increment by three only raises the price. Since $t \ge 3$, replacing any v by u strictly increases the Q price.

The full P cyclic derivatives have $\Delta u \in \{9, -4\}$ and

$$d_P = (9t+7, 9, 0)^T \text{ or } (-4t-3, -4, 0)^T.$$

The full Q derivatives are $(3,0,1)^T$ for phases $0,1$, and $(-5,0,-2)^T$ for phase 2. Define $v = (3-4t, -4, 2)^T$, which obeys $A_P v = 1$, and the authentic integral source transporter $T = I - v(A_P - A_Q)$. Direct multiplication gives

$$Td_Q = \begin{cases} (6-4t, -4, 3)^T, & j = 0,1, \\ (48t-41, 48, -26)^T, & j = 2. \end{cases}$$

The primitive parent integer kernel basis

$$K_P = \left((10+13t, 13, 0)^T,\ (3,0,1)^T\right)$$

then yields the four genuine discrepancy-coordinate types

$$(-1,3), \quad (0,3), \quad (3,-26), \quad (4,-26).$$

Since $\gcd(13,3) = 1$, every permitted pairing of source phases occurs in the true CRT period, and all four types occur. Transport into the physical child lattice with the source-frame-defined full basis

$$F = \begin{pmatrix} 29 & 4 \\ -4 & 0 \end{pmatrix},$$

whose determinant is 16 and whose columns satisfy the child congruence. More explicitly, with first-column insertion $j = 1$ and auxiliary index $j' = 2$ in the source-frame map $S = DB_t^{-T}U^{-T}B_t^T D$, direct multiplication gives $\mathrm{pr}_{u,v}(SK_P) = \left((29,-4)^T, (4,0)^T\right) = F$ for every $t \ge 3$; thus this basis is source-carried rather than chosen after observing the daughter optimum. The complete physical alphabet is exactly

$$\mathcal{D}_t = \{(-17,4), (12,0), (-17,-12), (12,-16)\}.$$

Its translation-invariant difference module is therefore

$$L_{\mathrm{diff},t} = \mathbb{Z}(29,-4) + \mathbb{Z}(0,16), \quad \det L_{\mathrm{diff},t} = 464,$$

whereas $\det \Lambda(C_t) = 16$. This proves the index 29 assertion without computing a daughter optimum. The child congruence $y \equiv 12x \pmod{16}$ is independent of t, because $q_C = 12 + 16t$ and $s_C = 47 \equiv 15 \pmod{16}$; the four physical letters are also independent of t as explicitly calculated above. Furthermore, the genuine letter $(-17, 4)$ has first coordinate not divisible by 29 and therefore lies outside $L_{\mathrm{diff},t}$, whose first coordinates are all multiples of 29. Since the quotient $\Lambda(C_t)/L_{\mathrm{diff},t}$ has prime order 29, the affine-origin theorem implies $L_{\mathrm{raw},t} = L_{\mathrm{diff},t} + \mathbb{Z}(-17,4) = \Lambda(C_t)$. Thus a single genuine absolute source letter repairs the full group-theoretic saturation defect, although finding the optimal price chamber still requires the separate certificate below.

It remains to prove the price minimizer assertion using the full source-carried basis, with no ray scan. The physical prices are

$$\Phi_t(z_A) = 336 + 48t, \quad \Phi_t(z_B) = 784 + 16t.$$

For $3 \le t < 14$, the pair (z_A, h_A) with $h_A = f_2 = (4, 0)$ is a full oriented determinant-16 basis. Every positive lattice vector has the form $w = jz_A + mh_A$ with $j, m \in \mathbb{Z}$. Since $w_y = 4j > 0$, necessarily $j \ge 1$. If $m \ge 0$, its price is no smaller than $\Phi_t(z_A)$. If $m = -l < 0$, then $w_x = 3j - 4l > 0$ forces $j \ge l + 1$, while

$$0 < \Phi_t(h_A) = 112 + 64t < 336 + 48t = \Phi_t(z_A),$$

so $\Phi_t(w) \ge (l+1)\Phi_t(z_A) - l\Phi_t(h_A) > \Phi_t(z_A)$. The first chamber therefore has exactly one optimum.

For $t \ge 14$, use instead $h_B = f_1 - 7f_2 = (1, -4)$. The pair (z_B, h_B) has determinant -16, and

$$\Phi_t(h_B) = 16t - 224 \ge 0, \qquad \Phi_t(z_B) - \Phi_t(h_B) = 1008 > 0.$$

Any positive $w = jz_B + mh_B$ has $w_x = j + m > 0$ and $w_y = 12j - 4m > 0$. If $m \ge 0$, the latter inequality implies $j \ge 1$, so $\Phi_t(w) \ge \Phi_t(z_B)$; equality away from $t = 14$ forces $j = 1, m = 0$. If $m = -l < 0$, the first inequality forces $j \ge l + 1$, so $\Phi_t(w) > \Phi_t(z_B)$. For $t > 14$ the unique optimum is z_B. At $t = 14$, $\Phi_t(h_B) = 0$; price equality requires $j = 1$, and positivity permits exactly $m = 0, 1, 2$, yielding $(1, 12), (2, 8), (3, 4)$. Since their first coordinates are $1, 2, 3$, the canonical lex-first point is $(1, 12)$. Every minimizing vector listed is primitive in the full Λ basis (its coefficients are $(1, m)$, or a basis vector). This proves the complete tie-wall classification. □

Remark 5.2. *The quotient obstruction in Theorem 3.1 applies only if the complementary direction is required to lie in L_{diff}. The source-derived full basis F has determinant* 16*, while the difference lattice has determinant* 464*; this explicit replacement of a non-saturated letter subgroup by the full carried integer basis is the constructive mechanism behind the repair. No unrestricted $P, Q \mapsto C$ optimal-WORD algorithm is asserted.*

Corollary 5.3 (Necessity of numerical price data)**.** *No deterministic rule that uses only the complete synchronized* unordered *physical CRT discrepancy alphabet and the full daughter congruence lattice, but ignores the positive price pair $(p + q, p + s)$, can return the correct globally minimizing positive child ray for all genuine positive unimodular Farey frames. This remains true even if the rule knows the translation-invariant and raw integer subgroup generated by the letter alphabet, their HNF and the full price-independent convex hull.*

Proof. For the genuine family of Theorem 5.1, all the listed inputs to such a rule remain exactly the same for every $t \ge 3$. In contrast the strictly positive true prices select $(3, 4)$ when $3 \le t \le 13$ and $(1, 12)$ when $t \ge 14$, with three ties at the separating wall $t = 14$. A rule using no numerical price data necessarily returns the same ray on all these frames and fails in at least one chamber. □

Remark 5.4. *The rule excluded by Corollary 5.3 is deliberately narrow. It may not use the actual positive price, other changing frame data, nor the ordered coordinates of the original complete parental optimal records. No impossibility for the unrestricted two-parent WORD transfer is claimed.*

Finite verification and scope

All universally quantified results are proved symbolically above; finite verification serves only as an independent consistency check. Exact-integer checks covered source/child unimodular transport in 960 genuine positive Farey-frame/flag cases and the determinant-ideal formula in 20,000 independently generated primitive lattice instances. The complete four-letter index-eight example was checked against separately optimized parent words. Further finite checks covered 25 odd parameters in each unbounded family and 37 parameters in the fixed-index family, comparing original parent optimizers and complete child residue prices with modular shortest-path calculations. These are the reported finite validation ranges, not exhaustive checks of every parameter or substitutes for the symbolic proofs. The rank-one parent-difference obstructions do not imply a no-go result for arbitrary source-aware transfer, which is not claimed here.

Contributions and novelty

The paper-specific contributions are the exact source-sized presentation of the synchronized parent-difference module, the quotient-order determinant obstruction in the genuine optimized-Farey setting, the two independently optimized infinite obstruction families, and the fixed-index family with an exact three-way price wall and its restricted price-blind selection impossibility. The underlying generalized CRT, lattice-index, Hermite/Smith normal form, and cyclic quotient arguments are classical and are not claimed as new in isolation.

Classical superimposition and synchronization of Christoffel words concern compatible word assemblies and discrete segments [1, 2]. More recent Cayley-seed synchronization investigates gap-free superposition of equal-length Christoffel conjugates [3]. Our optimized-parent input is a different, price-constrained target-indexed word category. Any broader priority comparison concerning the primitive daughter lattice, subgroup quotient, and essential new letters requires theorem-level comparison with that literature; complete external novelty certification is not asserted here.

Open problem and future work

It remains open in the scope of this work to develop a general, source-aware optimal-WORD transfer theorem from the complete *ordered* original parent records to the complete daughter optima, retaining the numerical daughter price data. The restricted impossibility in Corollary 5.3 applies only to selectors that omit the numerical price and use the specified unordered source alphabet and price-independent lattice data; it does not answer the unrestricted transfer question. A further investigation of source-aware transfer would therefore have to retain the distinction between complete optimum fibers and selected cyclic anchors.

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